Name	Type	Feature of Interest	Property	Characteristics	Components	Justification
The name of each IoT device	E.g., sensor, actuator, or device	What “thing” is being observed and/or acted upon?	Which aspect(s) of the “thing” is/are being observed and/or acted upon?	What are the characteristics of the device?	What are the components embedded in the device?	Why is the device selected for the proposed IoT solution?
Atlas Scientific Industrial RTD Temperature Kit (KIT-303)	Sensor	Water in each breeding tank	Water temperature	Class-A PT-1000 probe; approx. -55 to 220 °C; transmitter accuracy ±0.1 °C; 4–20 mA; 9–36 VDC. (Atlas Scientific, n.d.) https://atlas-scientific.com/transmitters/rtd-temperature-transmitter/	PT-1000 probe, RTD3 transmitter, immersion fitting, 24 VDC supply, Opta analogue input	Industrial current loop suits long fixed cable runs in a wet plant. The range exceeds the operating envelope and supports continuous local heater control.
Atlas Scientific Industrial pH Kit (KIT-102P)	Sensor	Water in each breeding tank	pH value	pH 0–14; stated accuracy ±0.002; internal PT1000; about 1 s response; 4–20 mA; 3 m cable. (Atlas Scientific, n.d.) https://atlas-scientific.com/kits/industrial-ph-kit/	Industrial pH probe and transmitter, immersion fitting, calibration solutions, 24 VDC supply	Provides a calibrated industrial interface for early detection of unsuitable acidity or alkalinity. Scheduled two/three-point calibration manages drift.
DFRobot Industrial RS485 Turbidity Sensor (SEN0710)	Sensor	Water in each breeding tank	Turbidity	0–1000 NTU; 0.1 NTU resolution; ±5% full scale; IP68; 10–30 VDC; RS-485 Modbus RTU. (DFRobot, n.d.) https://www.dfrobot.com/product-3104.html	Optical probe, Modbus interface, protective stilling tube, 24 VDC supply, shielded RS-485 trunk	A waterproof digital probe supports continuous clarity trends and helps identify filtration problems without noise-prone low-level analogue signals.
DFRobot Industrial Ammonia-Nitrogen and pH Sensor (SEN0711)	Sensor	Water in each breeding tank	Ammonia nitrogen, with pH and temperature compensation	NH3-N 0–100 mg/L; 0.01 mg/L resolution; ±3% full scale; pH 0–14; IP68; RS-485 Modbus RTU; 10–30 VDC. (DFRobot, n.d.) https://www.dfrobot.com/product-3107.html	Ammonia/pH/temperature probe, Modbus transmitter, immersion mount, calibration standards, 24 VDC supply	Direct ammonia indication addresses a high-risk water-quality condition. Compensation channels and local rate-of-change checks reduce false alarms.
DFRobot Submersible Water Level Sensor (KIT0139)	Sensor	Water in each breeding tank	Water depth and level	0–5 m range; ±0.5% full scale; 4–20 mA; 12–36 VDC; IP68 stainless probe. (DFRobot, n.d.) https://www.dfrobot.com/product-1863.html	Pressure probe, vented cable termination, stilling tube, 24 VDC supply, Opta analogue input	Independent level feedback prevents overflow, excessive drainage and dry-running, and verifies bounded water-change commands.
DFRobot Water Flow Sensor (SEN0217 / YF-S201)	Sensor	Tank inlet and drain pipes	Flow rate and transferred volume	1–30 L/min; Hall-effect pulse output; ±5% over 2–30 L/min; 3.5–12 VDC; 1/2-inch fitting. (DFRobot, n.d.) https://www.dfrobot.com/product-1517.html	Two sensors per tank, service unions, screened low-voltage cable, Opta pulse inputs	Inlet and drain totals cross-check level change and expose blocked pipes, pump faults or stuck valves while limiting exchanged volume.
DFRobot A02YYUW Waterproof Ultrasonic Sensor (SEN0311)	Sensor	Feed in each zone hopper	Remaining feed height	3–450 cm range; 3 cm blind zone; ±1 cm; 100 ms response; IP67; 3.3–5 V UART. (DFRobot, n.d.) https://www.dfrobot.com/product-1935.html	Top mount, dust shield, isolated UART interface, hopper-specific distance-to-fill calibration	One sensor per zone hopper provides a low-feed alert. Calibration accounts for the sloped pellet surface and converts distance to an approximate fill percentage.
Atlas Scientific EZO Humidity Probe (EZO-HUM-P-S)	Sensor	Air in each breeding/nesting zone	Relative humidity and air temperature	0–100% RH; stated ±2% RH from 5–95%; 3.3–5 V; UART/I2C; IP67 stainless probe. (Atlas Scientific, n.d.) https://atlas-scientific.com/probes/humidity-probe/	Shielded placement away from spray, isolated interface, mounting bracket, periodic comparison check	Zone humidity reveals condensation risk that can affect electronics and camera visibility, and provides context for nesting-area observations.

Name	Type	Feature of Interest	Property	Characteristics	Components	Justification
Aqua Medic Titanium Heater 500 W	Actuator	Water in each breeding tank	Heat input and water temperature	500 W; 110–240 VAC; intended for about 1000–1500 L; external controller required. (Aqua Medic, n.d.) https://www.aqua-medic.de/de/Produkte/Titanium-Heater~p3831	Titanium element, rated contactor, RCD, guarded mount, manual isolation, independent over-temperature cut-out	Corrosion-resistant construction suits aquaculture. PLC control uses conservative hysteresis, while the independent cut-out remains the final safety layer.
AQUAEL UNI PUMP 1500	Actuator	Tank recirculation and filtration line	Water circulation	1400 L/h; 19 W; maximum head 1.55 m; 220–240 VAC; IPX8; submerged or inline. (AQUAEL, n.d.) https://www.aquael.com/us/products/aquaristics-us/pumps-circulators-us/unipump/	Pump, strainer, check valve, service bypass, RCD, overload protection, rated contactor	Low-power continuous circulation supports filtration. Flow feedback detects loss of prime, blockage or pump failure.
Bürkert Type 6213 EV Normally Closed Valve, 24 VDC	Actuator	Tank inlet and drain lines	Water flow state	Two-way servo-assisted diaphragm valve; normally closed; 24 VDC option; water service; IP65 with suitable plug. (Bürkert, n.d.) https://www.burkert.com/en/products/solenoid-valves/general-purpose-2-2-solenoids/270895	Two valves per tank, selected DN/pressure variant, coil suppression, manual isolation, upstream strainer	Normally closed action fails safe on power loss. Paired with level and flow feedback, it enables bounded partial water exchange.
FIAP Electronic Feeder 10 kg with Spreader (1586-1)	Actuator / device	Feed supplied to turtles in each zone	Feeding time and estimated portion	10 kg hopper; 12 VDC; non-clogging screw; adjustable 2–12 m spread radius. (FIAP, n.d.) https://www.fiap.com/elektronikfuetterer-10-kg-mit-streuer.html	Hopper, screw and spreader, isolated enable relay, guarded chute, feed-level sensor, catch-test calibration	Commercial feeder reduces manual dosing. Runtime-to-mass calibration and hopper feedback provide a practical, auditable portion estimate.
Reolink RLC-811A	Sensor	Turtle activity, feeding and nesting areas	Video and event evidence	8 MP at 25 fps; H.264/H.265; 5× optical zoom; RTSP/HTTPS; IEEE 802.3af PoE; <12 W; IP67. (Reolink, n.d.) https://support.reolink.com/c/rlc-811a/	Image sensor, lens, processor, Ethernet/PoE interface, mount, privacy mask, local microSD fallback	Wired PoE avoids local adapters. Standard streams support local activity and nesting inference without continuous cloud video upload.